

in Mumbai, you can switch on your airconditioner 30 minutes before you reach home, dim the lights and have hot water ready for your bath well before you actually reach home.

The concept of IoT has been around for a long time. In the 1970s, some folks in the Computer Science department at Carnegie Mellon University decided to make things convenient for people by connecting the Coke vending machine in the department building to all the computers in the department, so that anyone from anywhere in the building could check if a cold bottle of Coke was available before going down to the vending machine, saving people from the unnecessary hassle of several futile trips to the machine. A simple convenient solution to the problem.



IPv4 used 32-bit addressing, while IPv6 uses 128 bits – an immense increase in the number of addresses, ushering the era of IoT

This is an isolated example, and there's no denying that large scale applications of IoT will naturally come with concerns of privacy and security of the enormous amounts of data that the devices would generate.

2. The reach

Potentially, IoT can connect everything – and that means everything that can be assigned an IP address – in the next few decades or even less. The underlying idea behind the internet as we know it is that each computer connected to it can be identified with a unique address – called the IP (Internet Protocol) address. The latest version of the Internet Protocol, IPv6, can support up to $2^{128} = 3.4 \times 10^{38}$ addresses, which is close to about one trillion trillion trillion addresses per computer on the planet!

This new protocol was developed because IPv4, currently the most common version, had a limited address space. Eventually the available